

# Registered structural geometry view specification

## Status

- Owner: Stage C0A3 registered structural-view integration
- Release target: mdstats 0.19.95a0
- Upstream contracts: C0A1 source-coordinate semantics and lattice gauge, C0A2 affine registration, Stage 11C1/C2 ring and tiling geometry, Stage 11C3 atom-resolved ring boundaries
- Downstream consumers: statistical-site association, registered density interpretation, structural classification, and visualization

## Purpose

A site-density field and an instantaneous framework geometry may inhabit different Cartesian coordinate measures. The package therefore requires one immutable, source-bound object that links:

1. the **physical structural geometry**, used for bond lengths, apertures, areas, volumes, and coordination; and
2. the **registered structural embedding**, used for density-space displacement and structural association.

The view must never silently replace physical geometry with registered geometry or mix coordinates from the two representations.

## Scope

C0A3 provides:

- framewise application of one resolved `FrameRegistrationResult` to persistent T and O ring atoms;
- registered ring, window, tile, and natural-tile/cage centers;
- registered tile-face vertices when a frame tiling catalog is supplied;
- reconstruction of orthonormal registered ring frames from transformed atom coordinates;
- explicit physical and registered geometry records linked by persistent atom and structural identities;
- periodic image reconstruction in the registered cell;
- trajectory-only orientation-continuity diagnostics with registration segment resets;
- deterministic serialization, replay, signatures, resource preflight, and fail-closed source validation.

C0A3 does **not**:

- discover statistical sites;
- compute M–O or M–T coordination fingerprints;
- modify C1/C2/C3 physical geometry;
- migrate MSD, density, or plotting consumers;
- infer temporal continuity for independent ensembles; or
- convert registered geometric sizes into physical bond or aperture claims.

## Coordinate convention

Row-vector coordinates are retained. For frame  $t$ , the resolved registration map is

$$\mathbf{q} = \mathbf{x}M_t + \mathbf{b}_t, \quad G_t = H_tM_t,$$

where  $\mathbf{x}$  is a physical Cartesian point,  $H_t$  is the physical cell in the resolved lattice gauge,  $\mathbf{q}$  is the registered Cartesian point, and  $G_t$  is the registered cell.

Every transformed structural point is produced by the exact C0A2 map. A ring atom with persistent source reference

$$(a, \mathbf{n}), \quad \mathbf{n} \in \mathbb{Z}^3,$$

is validated against the registered base atom and decomposed as

$$\mathbf{q}_{a,n} = \mathbf{q}_a + \mathbf{m}G_t, \quad \mathbf{m} \in \mathbb{Z}^3.$$

The integer registered image  $\mathbf{m}$  is solved and certified from the transformed coordinates. It is not assumed to equal the source image label when the source lattice basis has been reconciled by a unimodular gauge.

## Separation of physical and registered geometry

A `RegisteredRingStructuralView` contains two explicit children.

`PhysicalRingGeometrySnapshot`

physical T/O coordinates

physical center and orthonormal frame

physical aperture, area, perimeter, planarity, and T--O distances

`RegisteredRingEmbedding`

transformed T/O coordinates and registered images

reconstructed registered center and orthonormal frame

registered projected descriptors for association only

No generic center, distance, or aperture field is exposed at the combined level. Consumers must select physical or registered explicitly.

Physical values are copied without numerical reinterpretation from the compatible C2 frame geometry. Registered projected sizes are separately named and must not be reported as physical coordination geometry.

## Registered atom embedding

Each persistent T/O atom record retains:

- boundary kind and cyclic index;
- `RingAtomRef` source atom and image identity;
- element and Stage 11C3 chemical/environment annotations;
- physical Cartesian coordinates;
- registered Cartesian coordinates;
- registered unwrapped and wrapped fractional coordinates;
- registered integer image shift relative to the transformed base atom; and
- a periodic-image residual.

The transformed atom must agree with the registered base atom plus an integer registered-cell translation within `coordinate_tolerance`. A disagreement is a source-binding error, not a recoverable ring deformation.

## Registered ring-frame reconstruction

A non-rigid affine map does not preserve orthogonality. Therefore a registered local frame is reconstructed from transformed atoms; physical frame axes are never merely multiplied by  $M_t$  and relabeled orthonormal.

For transformed ordered oxygen coordinates  $\mathbf{q}_j$ :

1. compute the covariance about the vertex centroid;
2. take the least-variance eigenvector as a provisional plane normal;
3. sign the normal by the ordered polygon vector area;
4. project the polygon and compute its signed area centroid;
5. choose the in-plane  $\hat{\mathbf{u}}$  axis from the projected persistent cyclic-origin atom;
6. set

$$\hat{\mathbf{v}} = \hat{\mathbf{n}} \times \hat{\mathbf{u}};$$

7. verify unit length, mutual orthogonality, positive ordered area, and non-singular projected origin radius.

The reconstructed frame stores:

- center and orthonormal axes;
- covariance eigenvalues;
- projected area, perimeter, aperture witness, and planarity diagnostics;
- minimum projected atom radius;
- orthonormality error;
- transformed-physical-axis non-orthogonality diagnostic; and
- displacement between the reconstructed center and the directly transformed physical center.

The transformed physical axes may be retained only as a diagnostic. They are not the registered orthonormal frame.

## Orientation continuity

Persistent atom order fixes the frame gauge independently in every frame. For a trajectory, consecutive resolved frames also record

$$c_n(t) = \hat{\mathbf{n}}_{t-1} \cdot \hat{\mathbf{n}}_t, \quad c_u(t) = \hat{\mathbf{u}}_{t-1} \cdot \hat{\mathbf{u}}_t.$$

No continuity comparison is made:

- at the first selected frame;
- across a C0A2 `segment_reset_frame_indices` boundary; or
- for `FrameSemantics.ENSEMBLE`.

When enabled, a continuity dot below the declared threshold makes that ring view unresolved and records the failure. No sign flip is applied to hide the discontinuity.

## Tile, face, window, and cage embeddings

When a compatible `FrameTilingGeometryCatalog` is supplied, the view additionally records:

- natural-tile centers, explicitly designated as tile/cage centers;
- physical tile volume, surface area, diameter, sphericity, and orientation status unchanged;
- registered tile/cage center coordinates and periodic image data;
- transformed tile-face vertices and reconstructed face normals; and
- registered window centers linked to unchanged physical aperture and area metrics.

The natural tile is the current persistent cage-like structural unit. C0A3 does not assign a chemical cage name or a statistical-site identity.

## Status model

A frame is resolved, partial, or unresolved.

A ring view may be unresolved because:

- the C2 frame ring is unresolved;
- the C3 boundary is unresolved;
- C2 and C3 persistent identities disagree;
- transformed atom images cannot be certified;
- the transformed polygon is degenerate;
- the cyclic-origin direction is singular; or

- required trajectory orientation continuity fails.

Unresolved rings retain identity and a diagnostic but no partial registered geometry. Physical source geometry remains available through the upstream C2/C3 objects and their digests.

A missing or unresolved optional tiling frame does not erase valid ring views; it makes the frame partial and records a tiling diagnostic.

## Source binding and invariants

Construction fails before output when:

- frame indices or frame IDs disagree across C2, C3, registration, and the collection;
- the C3 catalog is not bound to the supplied C2 catalog;
- an optional tiling catalog contains duplicate or incompatible frame indices;
- a boundary atom identity or physical coordinate disagrees with C2 geometry;
- a transformed image does not close on the registered lattice; or
- resource estimates exceed the declared limits.

For every resolved ring:

- T/O identity order matches Stage 11C3 exactly;
- physical C2 metrics are retained exactly;
- the registered frame is right-handed and orthonormal;
- registered atom coordinates are exact C0A2 transforms;
- wrapped coordinate plus image shift reconstructs the registered coordinate; and
- no registered descriptor is used to overwrite a physical descriptor.

## Method provenance

The covariance-eigenvector best-fit plane follows the classical least-squares closest-plane construction introduced by Pearson [C0A3-1]. The implementation adapts that standard construction to an ordered periodic oxygen polygon, then uses the polygon's oriented vector area and its projected shoelace centroid as standard computational-geometry operations [C0A3-2]. The following parts are project-specific constructions rather than borrowed algorithms:

- the strict separation between physical and registered structural records;
- exact C0A2 affine-map application to persistent atom/image identities;
- independent certification of registered periodic images;
- reconstruction from transformed atoms instead of affine axis propagation;
- trajectory-segment orientation diagnostics; and
- source-bound canonical replay and resource contracts.

## Public API

```
build_registered_structural_geometry_view(
    collection,
    registration,
    frame_ring_geometry,
    ring_boundaries,
    *,
    frame_tiling_geometry=None,
    options=None,
    resources=None,
) -> RegisteredStructuralGeometryView
```

Primary persistent records:

LinkedStructuralAtomEmbedding  
PhysicalRingGeometrySnapshot  
RegisteredOrthonormalRingFrame  
RegisteredRingEmbedding  
RegisteredRingStructuralView  
RegisteredTileCageEmbedding  
RegisteredTileFaceEmbedding  
RegisteredWindowEmbedding  
RegisteredStructuralFrameView  
RegisteredStructuralGeometryView

## Serialization and signatures

The canonical schema is `mdstats.registered-structural-geometry-view.v1`.

The view digest binds:

- collection geometry/identity digest;
- registration signature;
- C2 frame-ring digest;
- C3 ring-boundary digest;
- optional frame-tiling digest;
- options and resources;
- all frame statuses, linked atom identities, physical snapshots, registered embeddings, and diagnostics.

`from_dict` performs source replay and requires byte-for-byte canonical payload equality. A modified payload, a changed source object, or a different registration result is rejected.

## Resource preflight

Before transforming coordinates, construction checks upper bounds on:

- frames;
- ring instances;
- linked T/O atom instances;
- tile instances;
- tile-face vertex instances; and
- window instances.

No partial catalog is returned after a resource failure.

## Acceptance tests

C0A3 is accepted when focused tests demonstrate:

1. identity registration reproduces registered T/O coordinates exactly while retaining all C3 identities;
2. a non-rigid shear/strain yields an orthonormal reconstructed frame even when affinely transformed physical axes are non-orthogonal;
3. physical bond lengths and aperture metrics remain byte-for-byte/numerically unchanged;
4. transformed ring atoms crossing periodic boundaries receive certified registered image shifts;
5. trajectory orientation checks honor segment resets and are skipped for ensembles;
6. unresolved C2/C3 rings remain explicit without partial registered records;
7. optional tile/cage, face, and window embeddings transform centers and vertices without overwriting physical metrics;
8. source mismatch, resource overflow, and serialization tampering fail closed; and
9. public analysis and root-package exports are stable.

## Next boundary

After C0A3, the next implementation stage is **C0B consumer migration**. Existing MSD, density, and plotting consumers are not migrated in this release.

## References

[C0A3-1] Pearson, K. (1901). *On Lines and Planes of Closest Fit to Systems of Points in Space*. Philosophical Magazine, Series 6, 2(11), 559–572. DOI: 10.1080/14786440109462720.

[C0A3-2] O'Rourke, J. (1998). *Computational Geometry in C*, 2nd edition. Cambridge University Press.